ITA03-MC Assignment 2

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MOBILE COMPUTING

ASSIGNMENT 2 – CO2

CO2 Coverage: Mobile Telecommunication Systems; Multiple Access Techniques (SDMA, TDMA, FDMA, CDMA, OFDMA); GSM Architecture, Services, Radio Interfaces, Localization, Calling, Handover & Security; Advanced GSM Extensions (GPRS, HSCSD, DECT).

Nature of Assignment: Individual **Duration:** 1 Week **Total Marks:** 50
Submission Date: _____

Q1 – Mobile Communication or Mobile Computing? 5 Marks

Problem Context – Field-Service Technology Audit

A utility company equips its field staff with: (a) a ruggedised tablet fetching work orders over cellular data, (b) a laptop plugged into Ethernet at different offices, (c) a fixed wireless router at a remote substation, (d) a push-to-talk handset, and (e) a tablet app that records meter readings offline and synchronises later.

Your Challenge

Investigate whether "wireless" and "mobile" mean the same thing, and whether mobile communication and mobile computing are the same thing.

Before analysis:

- Define at least four of your own classification criteria (e.g., user mobility, wireless link, tolerance of disconnection, location/context awareness, device resource limits).
- Predict how each of the five cases is classified (fixed-wired, nomadic, fixed-wireless, mobile) and whether it is mainly a mobile-communication or a mobile-computing problem, giving reasons.
- Design one additional ambiguous case of your own to test your criteria.

Test Your Reasoning

Build a small simulation of case (e) with link availability of 90%, 60% and 30%. Compare two strategies: send immediately (fail if disconnected) vs store-and-forward queue. Record updates delivered, delayed and lost, and mean delivery delay.

Inquiry

Which characteristics of mobile computing are exposed by your results that a wireless link alone does not solve? Which device categories need application-level changes?

Evidence Expected: Own classification criteria, reasoned predictions, an engineered ambiguous case, simulation counts for both strategies, and a justified conclusion.

Q2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic 6 Marks

Problem Context – Hostel-Campus Voice and Telemetry

A cell may use a 2 MHz band per direction. Option 1 is FDMA with 25 kHz channels. Option 2 is TDMA with 200 kHz carriers, each carrying 8 time slots. Both offer 80 nominal channels.

Your Challenge

Investigate whether equal nominal capacity means equal usable capacity.

- Estimate usable channels after guard bands, guard times, synchronisation and control overheads (state and justify your values).
- Predict which scheme gives better utilisation for (i) continuous voice, (ii) bursty data with about 10% duty cycle, and (iii) a 70/30 voice–data mix, and analyse why.

Design three contrasting traffic datasets, including one edge case of your own (e.g., very short calls or synchronised arrivals). Simulate dedicated-channel allocation under both schemes and record utilisation, blocking and wasted capacity.

Inquiry

Where do the two schemes diverge, and what causes it?

Design Decision

At a campus telemetry site, 90% of devices send a 200-byte report per minute. Is dedicated FDMA/TDMA channel allocation justified? Defend your answer.

Evidence Expected: Capacity calculations with overheads, reasoned predictions, three designed traffic conditions, utilisation/blocking evidence and a justified decision.

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem 7 Marks
Problem Context – Crowded Stadium Uplink

Several handsets transmit simultaneously in the same band to one base station, each spreading its data with its own code.

Your Challenge

Design a direct-sequence CDMA uplink simulation. Determine how data bits are spread, how the receiver despreads and decides, and what the processing gain is.

Before executing, predict the bit-error behaviour as the number of users grows for: (i) orthogonal (e.g., Walsh) codes with synchronous users, (ii) random non-orthogonal codes of the same length, and (iii) orthogonal codes where one user transmits at ten times the power. analyse your reasoning.

Implement and test with at least three code lengths (e.g., 8, 16, 64). Record bit errors over a stated number of bits, correlation values and processing gain.

Inquiry

- Why do orthogonal codes still fail when users are unsynchronised or powers are unequal?
- Why is power control essential in CDMA?
- Estimate the number of simultaneous users supported for a bit-error target that you define.

Evidence Expected: A working DS-CDMA simulation, predictions made before execution, bit-error evidence for three code lengths and three conditions, and a reasoned explanation of the results.

Q4 – OFDMA – Giving Each User the Subcarriers They Can Use Best 6 Marks

Problem Context – Lecture-Hall Downlink

Eight users share 64 subcarriers. Each user's channel quality varies across subcarriers (frequency-selective fading).

Your Challenge

- Before designing, determine how subcarrier spacing relates to symbol duration, why a cyclic prefix is needed, and verify orthogonality of at least three subcarriers numerically (FFT or inner product).
- Design three allocation approaches: fixed contiguous blocks, interleaved round-robin, and channel-aware best-subcarrier.

- Predict total throughput and fairness (e.g., Jain's index) of each, with reasons.

Implement using generated fading gains with fixed seeds over at least 1,000 channel realisations. Record sum throughput, per-user throughput and fairness.

Inquiry

Why can maximising total throughput starve some users? How does the number of users change the benefit of channel-aware allocation?

Design Decision

Select an allocation for a lecture hall in which every student must receive at least a minimum rate that you define and justify.

Evidence Expected: Numerical orthogonality verification, predictions, three allocation schemes implemented, throughput and fairness results, and a justified allocation choice.

Q5 – SDMA – Reusing the Same Channel in Different Directions 6 Marks

Problem Context – Highway Cell With Directional Antennas

Vehicles along a highway use the same frequency and time slot. The base station can serve them with an omni antenna, a 120° sector antenna, or an N-element uniform linear array steered toward each user.

Your Challenge

- Derive the array factor of an N-element uniform linear array and predict how main-lobe width and side-lobe level change with N.
- Define an SIR threshold and predict the minimum angular separation at which two users can share the same resource.
- Compute patterns for N = 2, 4, 8, 16 and evaluate the SIR at user B when the beam is steered at user A, for separations of 5°–60°. Record beamwidth, side-lobe level and the minimum separation meeting your threshold.

Inquiry

What limits SDMA in practice (angular spread, clustered users)? Why can two users in nearly the same direction not be separated?

Design Decision

At a toll plaza most vehicles arrive in the same direction. Would you rely on SDMA there? Defend your decision.

Evidence Expected: Array-pattern derivation, predictions, computed patterns and SIR results across angular separations, and a reasoned explanation of SDMA limits.

Q6 – Choosing Multiple-Access Techniques for Real Deployments 7 Marks

Problem Context – Four Deployments

Choose access methods for: (1) wide-area cellular voice, (2) a jam-resistant tactical link, (3) a high-rate stadium downlink, (4) a network of 5,000 low-rate sensors sending small reports.

Your Challenge

- Define at least five weighted criteria and justify the weights yourself.
- Rate each of SDMA, TDMA, FDMA, CDMA and OFDMA for each deployment using your own Q2–Q5 evidence wherever possible.
- Choose a primary technique and, where useful, a hybrid for each deployment.
- Perform a sensitivity check by changing weights by ±20%: does any choice change?

Inquiry

Why do real systems combine techniques (for example FDMA with TDMA)? Which of your choices is least robust, and why?

Evidence Expected: Independently justified criteria and weights, ratings traceable to your own Q2–Q5 evidence, sensitivity analysis and defended choices.

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate 13 Marks

Review the evidence you generated while working on Q1–Q6.

Select **ONE** solution that demonstrates a significant limitation as its problem size or operating requirement increases. Examples of limitations you may find in this variant:

- Dedicated FDMA/TDMA channels sit idle under bursty traffic.
- A CDMA link collapses under power imbalance or loss of synchronisation.
- Best-subcarrier OFDMA allocation starves weak users. ● SDMA fails when users are angularly clustered.

You are not permitted simply to replace the solution with a more advanced technology or standard and state that it is better. Your task is to understand the problem first and then develop an appropriate redesign.

A. Diagnose the Bottleneck 2 Marks

- Where most capacity, signalling or computational effort is being spent.
- Which counted quantity grows significantly as load or size increases.
- Whether resources are idle, wasted or repeated unnecessarily.
- Which characteristic of the original design causes the limitation.

Evidence Expected: A precise diagnosis supported by evidence from your original implementation and analysis.

B. Define the Design Objective 2 Marks

- State what your redesign must achieve before proposing it (e.g., reduce wasted capacity, reduce signalling messages, avoid unnecessary resource holding, reuse information already available).
- Specify a measurable objective wherever possible.

Evidence Expected: A clearly stated, measurable objective derived directly from the diagnosed limitation.

C. Design the Revised Strategy 4 Marks

- What you have changed and what property or information the new design exploits.
- Which operations or resource usages are avoided or reduced.
- Whether additional state, preprocessing or complexity is introduced, and any new trade-offs.
- Provide pseudocode, a flowchart or a message-sequence design.
- Marks are based on the quality of reasoning, not merely on naming a known technique.

Evidence Expected: A coherent redesign directly addressing the bottleneck, with a design representation and technical justification.

D. Predict Before Implementing 1 Mark

- Predict how the redesigned solution should differ from the original and in which measure you expect improvement.

Evidence Expected: A clear, testable prediction made before experimental validation.

E. Implement and Validate the Redesign 2 Marks

- Implement a prototype or simulation and compare Original → Redesigned under the same or equivalent test conditions.

- Use counted measures (e.g., slots used, messages, blocking, bit errors, throughput, fairness); run-time is supporting evidence only.

Evidence Expected: Comparable experimental evidence for the original and redesigned solutions.

F. Defend the Final Design 2 Marks

- Was the prediction supported, and what was the magnitude of improvement?
- What new costs or trade-offs appeared, and when might the original still be preferable?
- A redesign that does not outperform the original is not automatically unsuccessful if you correctly analyse why the hypothesis failed.

Evidence Expected: A critical, evidence-based evaluation and a defensible final engineering conclusion.

Mark Distribution

Question	Assessed Area	Marks
Q1	Mobile communication vs. mobile computing	5
Q2	FDMA and TDMA	6
Q3	CDMA	7
Q4	OFDMA	6
Q5	SDMA	6
Q6	Comparison of SDMA, TDMA, FDMA, CDMA, OFDMA	7
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	13
TOTAL		50

Submission Requirements (Digital)

- Assignment report (PDF) containing the required evidence for Q1–Q7. No handwritten work.
- Source code for all implementations/simulations (GitHub repository link with README and fixed random seeds).
- Raw experimental data (CSV) used to produce your results.
- Tables and graphs used in your analysis.
- Counted quantities (slots, messages, bits, blocked calls, codes, throughput) are primary experimental evidence. Execution time may be reported as supporting evidence but must not be the sole basis for conclusions.
- Standards parameters (frame durations, rates, header sizes, etc.) must be obtained from cited references; state assumptions wherever a value is assumed.

Assessment Verification

You may be asked to analyse any part of your design, code, test-data design, experimental results, predictions, analysis and Design Challenge in a brief verification

interpretion/viva. Disclose any AI or reference tools used. You must be able to analyse and defend the evidence as your own work.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook descriptions of mobile systems. It assesses your ability to:

Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate

Your conclusions and design decisions must be supported by your own reasoning and experimental evidence.

ANSWERS – MOBILE COMPUTING ASSIGNMENT 2 (CO2)

All results come from the Python simulation in sim2.py (fixed seeds, NumPy). Assumptions are stated with each answer; standards values (GSM slot structure) are from the GSM 05.xx specifications.

Q1 – Mobile Communication or Mobile Computing?

Criteria: C1 user moves while using the service; C2 wireless link; C3 must tolerate disconnection; C4 location/context awareness; C5 device resource limits (battery, storage). Wireless means no cable; mobile means the user or device moves. Mobile communication is about carrying the link during movement; mobile computing is about making applications work despite movement (disconnection, adaptation, synchronisation).

Case	C1	C2	C3	Class	Main problem	Reason
(a) Cellular tablet	Yes	Yes	Some	Mobile	Communication	Needs the link to stay up (handover, coverage)
(b) Laptop on Ethernet	Between sessions	No	No	Nomadic	Computing	Wired, so only re-addressing and re-configuration at each office
(c) Fixed wireless router	No	Yes	No	Fixedwireless	Communication	Wireless but no mobility
(d) Push-to-talk handset	Yes	Yes	Low	Mobile	Communication	Real-time voice over the radio link
(e) Offline meter app	Yes	Yes (intermittent)	High	Mobile	Computing	Must work offline, queue and synchronise later

Own ambiguous case: a truck-mounted tablet using the vehicle Wi-Fi hotspot, with GPS, that syncs at the depot. It is mobile and wireless (communication: hotspot/cellular

backhaul) and also needs offline GPS logging and sync (computing), so it is classified as both, with computing dominant because the depot is the only reliable link.

Simulation of case (e): 1,000 meter readings, one per time step, 1,200 steps, two-state link with mean outage 5 steps. Immediate = lost if the link is down. Queue = store-and-forward.

Link availability	Immediate: delivered / lost	Queue: on time / delayed / lost	Queue mean delay (steps)
90%	935 / 65	928 / 72 / 0	0.3
60%	588 / 412	525 / 475 / 0	2.8
30%	268 / 732	138 / 862 / 0	11.3

Inquiry: store-and-forward removes all losses but at 30% availability 86% of updates are delayed (mean 11.3 steps). A wireless link alone cannot give this: the application needs local storage, queueing, ordering, sync and conflict handling, and tolerance of delay. Devices needing application-level change: offline tablets/phones, sensors and wearables (small battery and storage); push-to-talk and fixed routers need none.

Q2 – FDMA vs TDMA Under Different Traffic

Usable channels: FDMA: 2 MHz / 25 kHz = 80; minus 2 guard channels (band edges) and 1 control channel = **77**. TDMA: 10 carriers × 8 = 80; minus 1 carrier as guard band = 9 carriers (72 slots); minus 2 slots for control = **70**. TDMA slot efficiency: a GSM slot is 156.25 bits, of which 114 carry data (tail, training sequence, guard time are overhead), so 114/156.25 = **73%** payload. FDMA has no slot overhead (100% assumed). So equal nominal capacity (80) gives 77 vs 70 channels, and 77 vs 51 payload-equivalent channels.

Prediction: continuous voice – FDMA better (no slot overhead); bursty 10% data – both poor, because the dedicated channel idles 90% of the time; 70/30 mix – FDMA better but both wasteful.

Simulation: dedicated allocation, Poisson arrivals 0.5 calls/s (edge: 20 calls/s), mean hold 120 s, 20,000 s. Utilisation = useful bits / capacity; Held = channels reserved.

Traffic dataset	FDMA block / util / held	TDMA block / util / held
Voice (duty 100%)	0.6% / 77.8% / 77.8%	2.2% / 61.7% / 84.5%
Bursty data (duty 10%)	0.6% / 7.8% / 77.8%	2.2% / 6.2% / 84.5%

70/30 mix	0.6% / 56.8% / 77.8%	2.2% / 45.1% / 84.5%
Edge: very short calls (3 s, 1 s setup)	10.8% / 69.6% / 92.8%	17.3% / 51.8% / 94.6%

Where they diverge: FDMA is better in all four sets, because of TDMA overheads (fewer channels and 73% payload), which is largest for voice. Both schemes diverge from ideal on bursty data: channels are held 78–85% of the time but carry data only 6–8% of the time, so about 90% of the reserved capacity is wasted, caused by dedicated allocation, not by the scheme. In the edge case (3 s calls, 1 s setup) setup time is held but carries no data, so blocking rises to 11% and 17%.

Design decision (telemetry, 200 B per minute): dedicated channels are **not justified**. A 200 B report at 9.6 kbps takes about 0.17 s per minute, a duty cycle of about 0.3%, so more than 99% of a dedicated channel would be idle. Use a shared random-access or reservation channel.

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem

Design: each user bit b (±1) is multiplied by its code c (L chips, ±1) to give L chips per bit. The receiver multiplies the sum of all signals by the wanted code and sums over L chips (despreading), then decides by the sign. Processing gain = chip rate / bit rate = L (8 → 9.0 dB, 16 → 12.0 dB, 64 → 18.1 dB).

Predictions before running: (i) Walsh, synchronous – no multi-access errors for any number of users up to L; (ii) random codes – MAI grows with users, errors increase; larger L helps; (iii) Walsh with unequal power – ideal orthogonality survives only if synchronised; with chip offsets the 10× user leaks into the wanted user, so errors appear.

Test: K = L/2 users, 20,000 bits for user 1, Eb/N0 = 10 dB (noise alone gives BER about 4×10^{-6}), seed 3. Condition (iii) = Walsh, random chip offset for the interferers, user 2 at 10× power.

L	Users	Processing gain	(i) Walsh sync: errors (mean |corr|)	(ii) Random: errors (mean |corr|)	(iii) Walsh async + 10× power: errors
8	4	8 (9.0 dB)	0 (0.0)	1332 (0.333)	1135
16	8	16 (12.0 dB)	0 (0.0)	676 (0.165)	34

64	32	64 (18.1 dB)	0 (0.0)	2379 (0.104)	9

Results match the predictions: (i) is 0 errors with zero cross-correlation; (ii) BER 6.7%, 3.4%, 11.9% at K = L/2 (more users and cross-correlation about $1/\sqrt{L}$); (iii) 5.7%, 0.17%, 0.045%, so a larger L reduces but does not remove the damage.

- **Why orthogonal codes still fail:** orthogonality holds only at zero time offset. With offset, cross-correlation is not zero, and a 10× stronger interferer adds 10× more interference, so even small non-zero correlation produces errors.
- **Why power control is essential:** all users share the band, so a nearby strong user swamps far users (near-far problem). Equal received power keeps MAI equal for everyone and maximises users.
- **Users supported (BER ≤ 1%, random codes, Eb/N0 10 dB):** L = 8 → 2 users, L = 16 → 5, L = 64 → 11. Capacity grows with processing gain but is far below L because of non-orthogonal interference.

Q4 – OFDMA

Basics: subcarrier spacing $\Delta f = 1/T_{sym}$, so a longer symbol gives closer subcarriers. A cyclic prefix (a copy of the end of the symbol put in front, longer than the delay spread) absorbs multipath echoes, so symbols do not overlap (no ISI) and subcarriers stay orthogonal. **Orthogonality check** (inner product of 64-sample symbol, normalised): ⟨1,2⟩ = 0.000000, ⟨1,3⟩ = 0.000000, ⟨2,3⟩ = 0.000000, ⟨3,3⟩ = 1.0.

Predictions: block and interleaved give the same sum rate and the same low fairness (user averages differ); best-subcarrier gives the highest sum rate and the lowest fairness, because strong users win every subcarrier; a channel-aware scheme with fairness (proportional-fair) gives lower sum but better fairness.

Test: 8 users, 64 subcarriers, Rayleigh fading, mean SNR from 20 dB (user 1) to 2 dB (user 8), rate log2(1+SNR), 1,000 realisations, seed 11. Rates in bits per OFDM symbol.

Scheme	**Sum rate**	**User 1 … User 8**	**Jain index**	**Mean min-user rate**	**5th-percentile min-user rate**
Fixed blocks	212.5	47.0 / 40.6 / 34.4 / 28.1 / 22.6 / 17.4 / 13.1 / 9.2	0.806	9.2	5.8

Interleaved	213.1	47.5 / 40.7 / 34.6 / 28.1 / 22.4 / 17.6 / 13.0 / 9.3	0.805	9.3	5.9
Best subcarrier	424.1	247.7 / 116.0 / 44.3 / 12.9 / 2.7 / 0.4 / 0.0 / 0.0	0.29	0.0	0.0
Channel-aware proportionalfair	289.3	47.9 / 47.0 / 45.3 / 41.8 / 36.3 / 29.8 / 23.4 / 17.7	0.913	17.7	14.2

- **Why max throughput starves users:** the best scheme gives each subcarrier to the user with the highest rate, so low-SNR users 6–8 get about 0 (Jain 0.29), though total is the highest (424 against 213).
- **Effect of number of users** (equal mean SNR, best against interleaved sum rate): 2 users 1.26×, 4 users 1.46×, 8 users 1.61×, 16 users 1.73×. More users give more choice per subcarrier (multiuser diversity), so the gain of channel-aware allocation rises.

Design decision (lecture hall): define minimum rate = 15 bits per symbol per student (assumed; enough for a basic lecture stream). Best subcarrier gives 0 to the weak users, and blocks/interleaved give the weakest user 9.3 (fails), so choose **channel-aware proportional-fair**: weakest user 17.7 on average (5th percentile 14.2, just under 15), sum rate 289 (36% above interleaved), Jain 0.91. If a strict guarantee is needed, add a minimum-rate reservation to the weakest users first.

Q5 – SDMA

Array factor (N elements, spacing d = λ/2, steered to θ_o): AF(θ) = Σ e^{jn ψ}, ψ = π(sin θ − sin θ_o), so |AF| = |sin(Nψ/2)| / (N |sin(ψ/2)|). Nulls at Nψ/2 = kπ. **Prediction:** main-lobe width falls as about 2/N (in sin θ); first side lobe stays about −13 dB for large N, so a larger N gives narrower beams, not lower side lobes. **SIR threshold:** 10 dB (assumed). Two users share a resource if SIR at A (interference from B) is at least 10 dB, i.e. $|AF(\Delta\theta)|^2 \leq -10$ dB.

N	**Half-power beamwidth**	**First null**	**First side lobe**	**SIR at 5° / 10° / 20° / 45° (dB)**	**Min separation for SIR ≥ 10 dB**

2	60.0°	90.0°	none (no side lobe)	0.1 / 0.3 / 1.3 / 7.1	53°
4	26.3°	30.0°	-11.3 dB	0.4 / 1.7 / 7.8 / 11.4	23°
8	12.8°	14.5°	-12.8 dB	1.8 / 8.4 / 13.0 / 22.9	11°
16	6.4°	7.2°	-13.1 dB	8.6 / 13.2 / 20.9 / 24.2	6°

(Users A at 0°, B at separation $\Delta\theta$, equal power and distance; sample separations shown, computed for 5°–60° in 1° steps. Exact nulls give very high SIR, e.g. N = 4 at 30°.) Omni antenna: all users on the same resource, SIR = 0 dB, so none can share. 120° sector: users in different sectors can share, users in the same sector cannot.

- **Limits in practice:** multipath angular spread widens the apparent direction, clustered users cannot be separated, and imperfect calibration, mobility and array cost limit gain. Two users in nearly the same direction lie inside one main lobe (beamwidth 6.4° even for N = 16), so the array cannot give one a high gain and the other a low one.

Design decision (toll plaza): do **not** rely on SDMA. Vehicles arrive from the same direction, so angular separation is a few degrees, below the 6–23° needed. Use TDMA/OFDMA scheduling there, with SDMA only as an extra gain.

Q6 – Choosing Multiple-Access Techniques

Criteria and rating (1 to 5, from Q2–Q5 evidence): spectral efficiency (OFDMA 5: Q4 gain; FDMA 2: Q2 idle waste), interference/jam resistance (CDMA 5: processing gain, Q3), simplicity/cost (FDMA 5, others 2–3), bursty and low-rate suitability (FDMA 1: Q2 waste; OFDMA 4: shared subcarriers), scalability (CDMA/OFDMA 4: Q3, Q4). Ratings: SDMA 4/3/2/3/3, TDMA 3/2/3/3/3, FDMA 2/2/5/1/2, CDMA 4/5/2/3/4, OFDMA 5/3/2/4/4. Weights are chosen by the deployment need.

Deployment (weights: efficiency / jam / simplicity / bursty / scale)	**SDMA**	**TDMA**	**FDMA**	**CDMA**	**OFDMA**	**Primary (hybrid)**
Cellular voice (.30/.15/.15/.10/.30)	3.15	2.85	2.35	3.75	3.85	OFDMA (FDMA+TDMA proven alternative; SDMA sectors)

Tactical link (.10/.50/.10/.10/.20)	3.0	2.5	2.2	4.2	3.4	CDMA (+ frequency hopping)
Stadium downlink (.45/.10/.05/.10/.30)	3.4	2.9	2.05	3.9	4.25	OFDMA + SDMA
5,000 sensors (.05/.10/.30/.35/.20)	2.75	2.9	2.55	3.15	3.35	OFDMA / random-access TDMA on narrow FDMA bands

Sensitivity (each weight ±20%, renormalised): no primary choice changed in any deployment. Margins: tactical 0.80 (CDMA over OFDMA), stadium 0.35, sensors 0.20 (OFDMA over CDMA), cellular voice 0.10 (OFDMA over CDMA).

Why real systems combine techniques: one technique does not fit every constraint: FDMA splits the band into carriers (isolation, simple), TDMA shares each carrier among users (more users per carrier), SDMA adds reuse by direction. GSM = FDMA + TDMA. **Least robust choice:** cellular voice, margin 0.10, so a small change in rating or in the weight for simplicity (where FDMA/TDMA is cheap and proven for voice) could flip it; the sensor choice is also close and needs a random-access element.

Q7 – Diagnose, Redesign and Validate (FDMA/TDMA dedicated channels under bursty traffic)

A. Diagnosis (2)

Most capacity is spent holding channels, not carrying data. In Q2 channels were held 78–85% of the time but carried data 6–8% (bursty). In the telemetry case a report needs 0.17 s but the dedicated channel is held for setup 1 s + 0.17 s + release timer 3 s = 4.17 s, so 96% of the held time is idle or overhead. The counted quantity that grows is channel-seconds held (and blocked calls), proportional to the number of devices: at 1,200 devices the original blocked 20.8%. **Cause:** the design reserves a channel per session, regardless of the actual data needed.

B. Objective (2)

Hold a channel only while data is actually being sent. Targets (measurable): held channel time below 10% at 1,200 devices (original 95%), blocking below 1% at 3,000 devices (original 67%), mean added wait below 0.5 s, with the same 70 channels.

C. Revised strategy (4)

Change: replace per-session dedicated allocation by **burst-level demand assignment**. A device sends a short request in a contention mini-slot and gets a channel only for its burst (0.17 s + 0.03 s overhead, assumed to include about 15% retries). The channel returns to the pool immediately (no setup and no release timer). If no channel is free the

request waits in a queue for up to 5 s. **Property used:** reports are short, rare (one per minute) and delay-tolerant. **Avoided:** call setup and idle holding (4.17 s → 0.2 s per report). **New costs:** contention-channel capacity, collisions and retries, queue state at the base station, a latency up to 5 s, extra scheduler complexity.

```
ALGORITHM BurstAssign(Device, Report)
BEGIN
  SEND Request TO BaseStation on ContentionSlot
  IF collision THEN WAIT random backoff ; RETRY END IF
  IF FreeChannel exists THEN
    Grant ← FreeChannel
  ELSE
    ENQUEUE Request ; START TIMER 5 s
    IF TIMER EXPIRES THEN DROP Request ; RETURN BLOCKED END IF
  END IF
  SEND Report ON Grant
  RELEASE Grant TO FreeChannelPool
  RETURN DELIVERED
END
```

D. Prediction (1)

Channel-held time falls by more than 90% (4.17 s → 0.2 s per report); blocking falls to about 0% even at 3,000 devices; useful utilisation rises because blocked reports now succeed; the waiting time stays near 0 (offered load 3,000 × 0.2/60 = 10 Erlang, much below 70).

E. Validation (2) – 70 channels, 3,600 s, Poisson reports (1 per minute per device), seed 5

Devices	Reports	Original: delivered / blocked (%) / held / useful util	Redesign: delivered / blocked / held / useful util	Redesign mean wait
300	18308	18308 / 0 (0.0%) / 30.3% / 1.24%	18308 / 0 / 1.5% / 1.24%	0.0 s
1200	72858	57682 / 15176 (20.8%) / 95.4% / 3.89%	72858 / 0 / 5.8% / 4.92%	0.0 s
3000	181818	60015 / 121803 (67.0%) / 99.3% / 4.05%	181818 / 0 / 14.4% / 12.27%	0.0 s

F. Defence (2)

Prediction supported. Held channel time fell from 95.4% to 5.8% (1,200 devices) and from 99.3% to 14.4% (3,000), blocked reports fell from 20.8% and 67% to 0%, and useful utilisation rose from 4.05% to 12.27% (3,000). At 300 devices both deliver all reports (useful utilisation equal at 1.24%), but the redesign holds 1.5% against 30.3%, so the gain there is freed capacity only. **Trade-offs:** the simulation assumes the contention overhead is fixed at 0.03 s (collisions and retries not modelled in detail), a contention channel must be reserved, and latency can reach 5 s under overload. **Original still preferable** for continuous voice or long data sessions (Q2: FDMA 77.8% utilisation), where the channel is busy all the time and setup overhead is small.

Disclosure: the simulations and analysis were prepared with an AI tool; code is in sim2.py with fixed seeds and should be reviewed and run by the student before submission.